

Stress Prediction from Wrist Physiological Signals using WESAD

Biddiscombe, Daans, Kubišová, Sedra

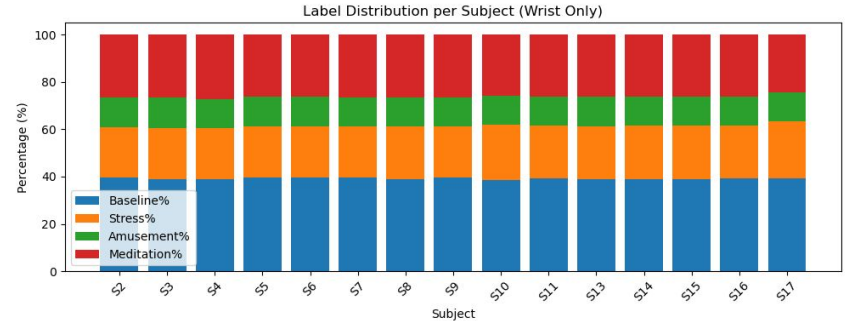


Problem and Dataset

Can we classify affective states from wrist signals – both 4-class and binary stress detection?

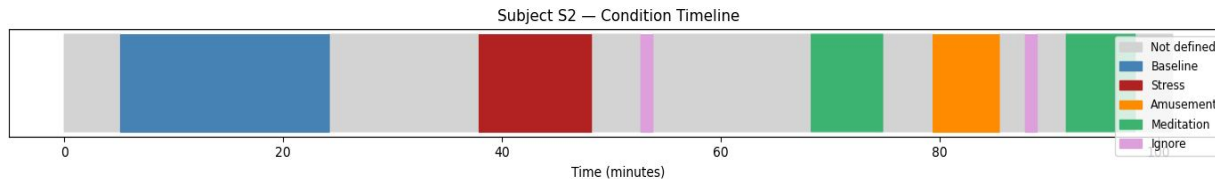
WESAD Dataset

- 15 subjects, controlled lab setting
- 2 devices recorded: chest (RespiBAN) + wrist (Empatica E4)
- We use 4 signals (wrist only) :
 - BVP (64 Hz)
 - EDA (4 Hz)
 - TEMP (4 Hz)
 - ACC (32 Hz)
- 4 conditions:
 - Baseline (rest)
 - Stress (TSST)
 - Amusement (videos)
 - **Meditation ← our extension**



	Paper (Schmidt et al., 2018)	Our Work
Classes	3 (baseline, stress, amusement)	4 (+ meditation)
Models	RF, AdaBoost, LDA, DT, kNN	LR, Linear SVM, XGBoost, MLP
Wrist accuracy	76% (3-class) · 88% (binary) (RF/AdaBoost)	To Be Discussed

Data Preprocessing

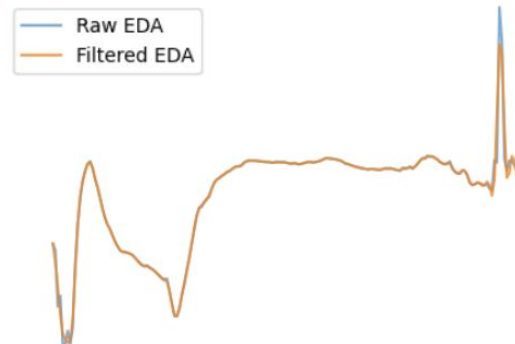


*Our 4 raw signals still had sensor noise, drift, and motion artifacts → **clean each signal***

BVP → **Bandpass filter 0.5 – 8 Hz** to keep the cardiac band only
(removing baseline wander and high frequency noise)

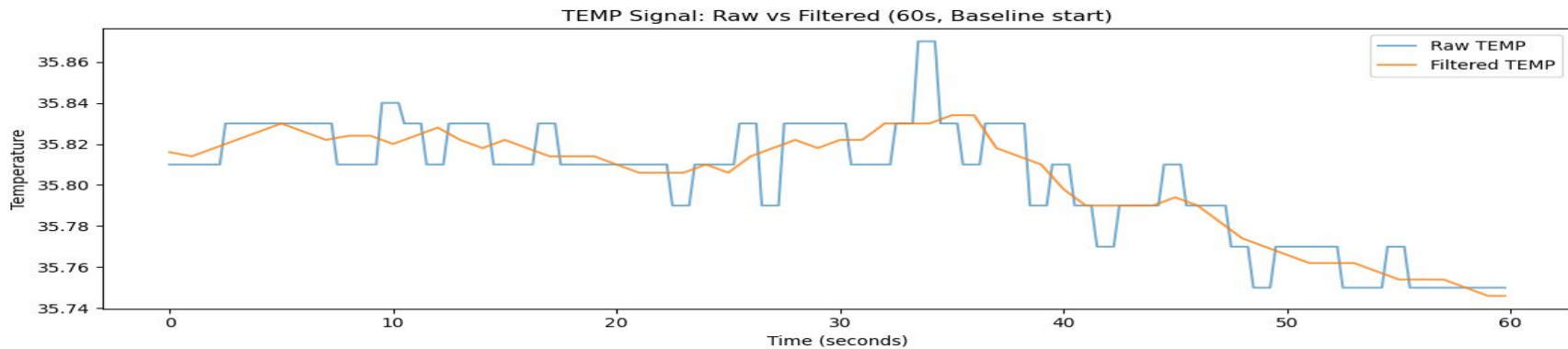


EDA → **Low-pass at 1 Hz** to smooth jitter
(raw is already slow signal at 4 Hz)

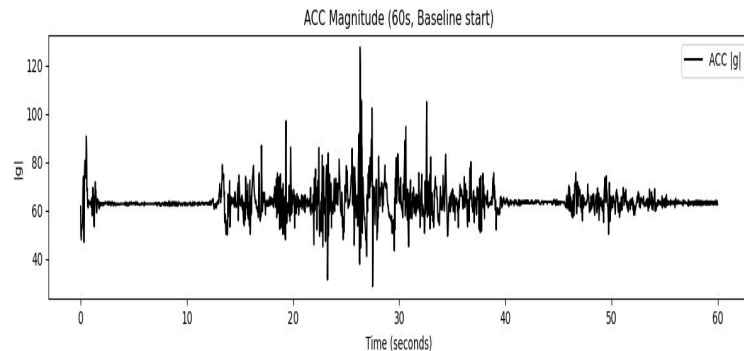
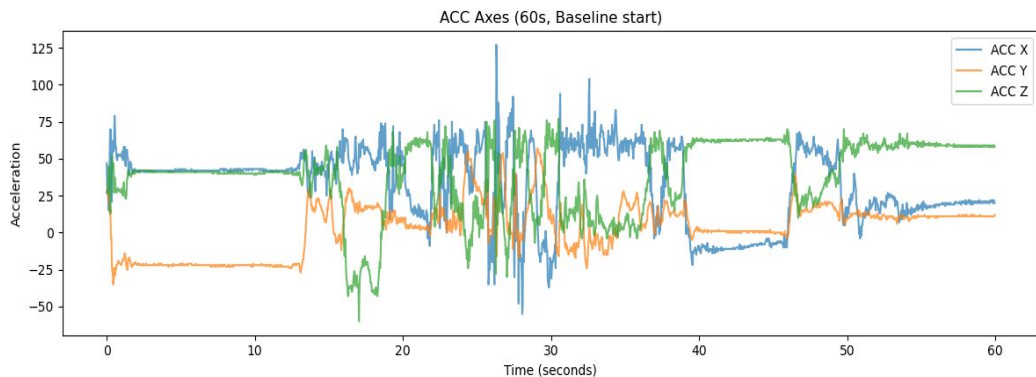


TEMP → 5-second moving average to smooth steps without killing trends (smoother)

Data Preprocessing



ACC → keep raw axes, but computing magnitude so movement is orientation-invariant across subjects
(smooth magnitude with a 1-s moving average to reduce jitter)



Feature Engineering

From our 4 different signals we extracted **41 different features** that we considered useful to predicting our target.

We mirrored the features extracted from the paper excluding chest data (only wrist):

- BVP (12): mean/std/min/max/range/slope heart rate + HRV time/freq metrics
- EDA (9): mean/std/min/max/range/slope + phasic SCR peaks to capture sympathetic arousal
- TEMP (6): mean/std/min/max/range/slope
- ACC (14): mean/std/min/max/range/slope, axis stats, magnitude stats, movement energy, peak motion freqs

2 quality flags mark missing SCR or unreliable HRV freq bands

**Segmentation: 60s windows,
0.25s step → 179,817 windows**

Feature Selection

Two-step process

Step 1: Correlation Clustering

- 8 pairs with correlation > 0.95 identified (e.g. temp_mean/min/max, eda_mean/min/max, bvp_hrv_std/rmssd)
- Hierarchical clustering at threshold $|r| \geq 0.90$
- 41 features → **33 cluster representatives**
- One representative per cluster: highest variance wins

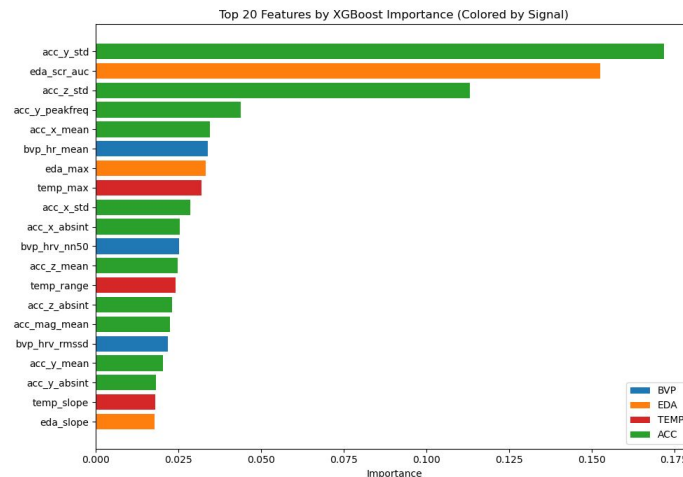
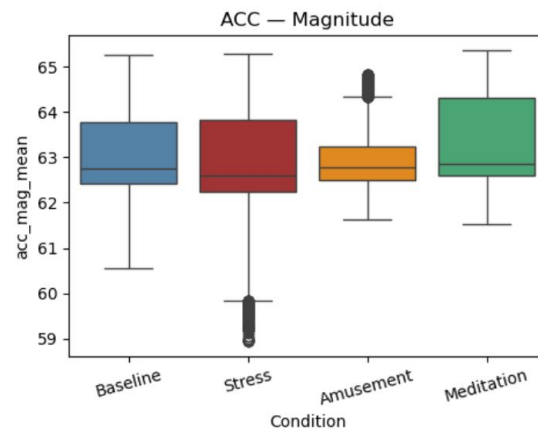
Step 2: XGBoost Importance Ranking

- XGBoost fit on all 33 representatives
- Top 20 selected by importance score

⚠ **ACC dominates: 9/20 features are accelerometer**

- *Despite ACC showing almost no class separation in the boxplots*
- *Possible reason: posture/subject identity rather than stress physiology*

⚠ **Limitation: feature selection done outside LOSO loop**



Modelling — LOSO Cross-Validation Framework

Why LOSO?

- 179,817 windows from 15 subjects — random splits invalid (same subject in train and test)
- Hold out 1 subject entirely per fold → 15 folds, true subject-independent evaluation
- StandardScaler fit on training subjects only per fold

4 Models — fixed hyperparameters (literature values, not tuned inside LOSO):

- Logistic Regression — linear baseline
- Linear SVM — large-margin linear classifier, scales to large datasets
- XGBoost — gradient boosted trees
- MLP — shallow neural network (100, 50 units)

2 Feature sets: Full (41) vs Selected (20)

Primary metric: Macro F1

- Accuracy misleading — baseline is 39% of data

Multi-Class Results (4-class)

Model	F1 (Full)	Acc (Full)	F1 (Selected)	Acc (Selected)
Logistic Regression	0.524	0.576	0.506	0.564
XGBoost	0.514	0.625	0.519	0.629
Linear SVM	0.507	0.651	0.487	0.631
MLP	0.478	0.573	0.434	0.533

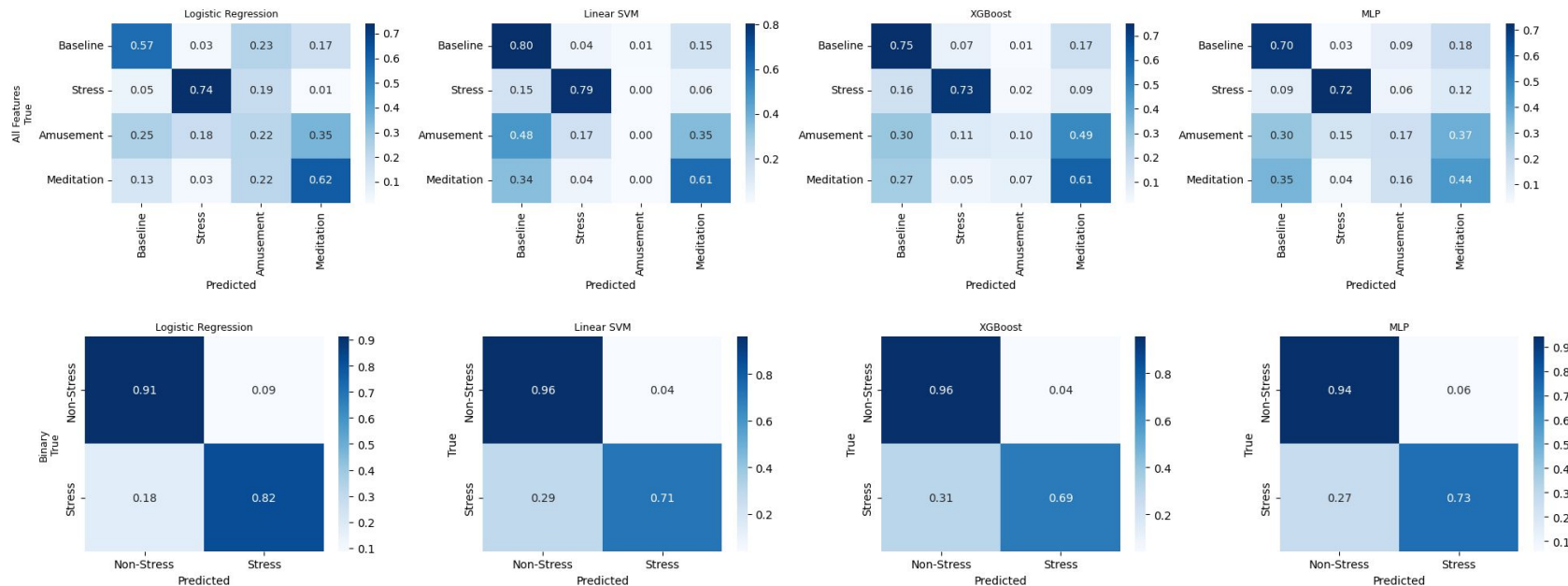
- Best macro F1: **Logistic Regression 0.524** — full features
- Linear SVM has highest accuracy (0.651) but lower F1 → predicts majority classes (baseline/meditation), misses amusement
- Feature selection hurt 3/4 models — full feature set preferred
- Only XGBoost marginally benefited from selection (+0.004)

Binary Results (Stress vs Non-Stress)

Model	F1 (Full)	Acc (Full)	F1 (Selected)	Acc (Selected)
Logistic Regression	0.846	0.891	0.839	0.882
Linear SVM	0.841	0.905	0.810	0.883
XGBoost	0.833	0.898	0.845	0.907
MLP	0.828	0.894	0.777	0.861

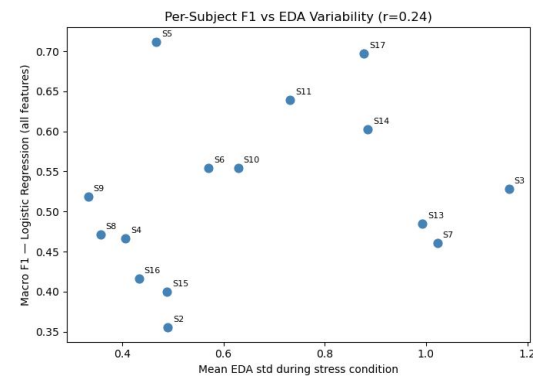
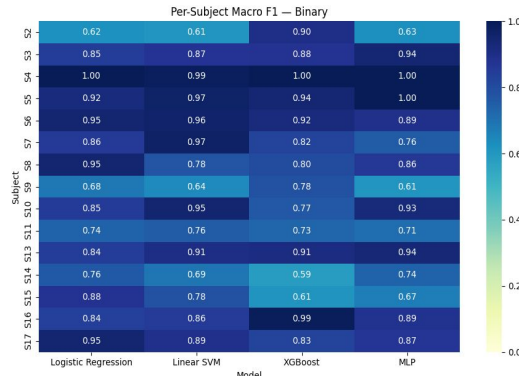
- Best macro F1: **Logistic Regression 0.846** — full features
- All models above 0.77 F1 — much stronger than multi-class
- Feature selection again hurt 3/4 models — full features preferred
- XGBoost is again the only model that benefits from feature selection
- Binary is dramatically easier than 4-class — +0.32 F1 average gain

Confusion Matrices



- Stress: consistently well classified (0.69 - 0.82 diagonal) for both binary and multi-class
- Baseline: reasonable classification (0.56 - 0.80)
- Amusement: near-zero diagonal — systematically predicted as Baseline or Meditation → a signal failure

Per-Subject Variability



- **Multi-class F1 range: 0.25–0.71** — large subject-to-subject variability
- **Binary: 0.59–1.00** — tighter and consistently strong across all subjects
- Best subjects: S5, S14, S17 · Worst: S2, S15, S16 → consistent across models
- EDA variability during stress vs F1: $r=0.24$ — weak, does not explain performance

Individual physiology drives performance more than signal strength — some subjects are simply harder to generalize to

Benchmark, Critical Analysis & Conclusions

Task	Our Best	Our F1	Our Accuracy	Paper (RF/AB)
4-class	LR (full)	0.524	57.6%	76% (3-class)
Binary	LR (full)	0.846	89.1%	88% ← we beat it

Critical Analysis:

- **Multi-class gap expected:** harder task (4 vs 3 classes), no RF, fixed hyperparameters
- **Bias:** 15 subjects, 12M/3F, students only
- **ACC dominance:** feature importance may reflect posture, not stress physiology
- **GDPR Article 9:** sensitive health data, workplace misuse risk

Conclusions:

- **Binary stress detection works well** — beats paper benchmark
- **4-class is much harder** — amusement effectively undetectable at the wrist (AUC=0.54)
- **Meditation** as 4th class is **meaningful** but increases confusion between relaxed states

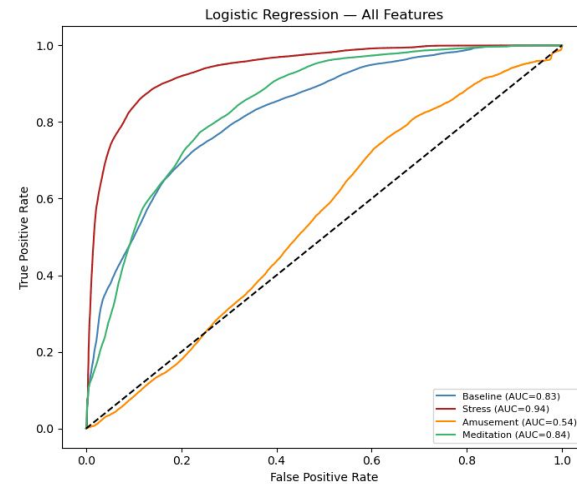
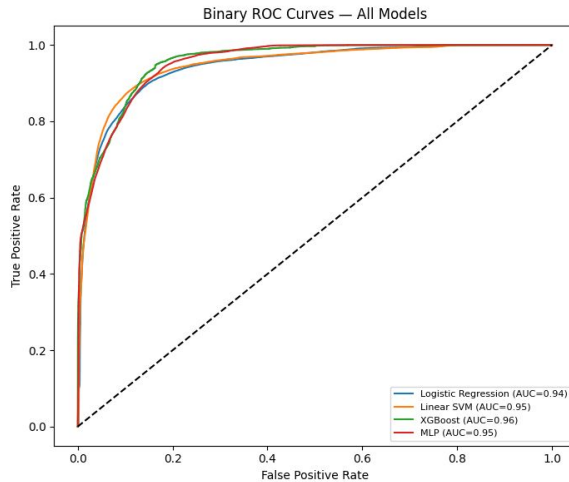
***Future Work:** LOSO-internal hyperparameter tuning · 1D CNN/LSTM on raw signals · larger, more diverse dataset*



Thank you for your attention
Any questions?



ROC Curves



- **Binary:** all models **AUC > 0.93** — excellent discrimination
- **Multi-class:**
 - Stress **AUC=0.94**, Meditation **AUC=0.84**, Baseline **AUC=0.83**
 - Amusement **AUC=0.54** — random chance → physiologically indistinguishable at the wrist regardless of model or task formulation